AAE 450 – Spacecraft Design - Spring 2026 Syllabus

Applicable to All Sections

General Course Information

- **Course number and title**: AAE 45000 Spacecraft Design, lecture sections 025, 026 and 030 and lab sections 029, 028 and 031
- **CRNs**: 23126, 23127, 35941 (lec. 025, 026, 030); 23129, 23130, 35942 (labs 028, 029, 031)
- **Meeting days, times and locations**
  - **Section 026 Lecture:** Tuesdays & Thursdays, 9:30 – 10:20 AM, STEW 320
  - **Section 028 Lab Presentations:** Thursdays, 1:30 – 4:20 PM, ME G061
  - **Section 025 Lecture:** Tuesdays & Thursdays, 11:30 AM – 12:20 PM, CRTN 1011
  - **Section 029 Lab Presentations:** Tuesdays, 1:30 – 4:20 PM, SMTH 118
  - **Section 030 Lecture:** Mondays & Wednesdays, 2:30 – 3:20 PM, WALC 1087
  - **Section 031 Lab Presentations:** Tuesdays, 10:30 AM – 1:20 PM, ME 2061
- **Course credit hours**: 3.0
- **Prerequisite**: Senior standing in AAE

Course Description

Senior students perform a team-based spacecraft design, requiring application of the education and skills developed in the aerospace curriculum. Components include analysis methods for preliminary design, development of an initial vehicle concept, and development of a complete numerical model of the mission, culminating in oral and written reports by the teams.

Additional Comments:

- There will be an emphasis on learning methods and techniques of the design process, using current best practices, in addition to the application of technical skills acquired earlier in the AAE curriculum.
- Systems engineering, project management, cost estimation and mission assurance are important aspects of the course.
- Brief overviews will be provided for specific technical areas required to perform the design of a space system. The emphasis will be on topics *not presently covered in the undergraduate curriculum*.
- The project will be a complete space system, including ground support, mission operations and science product generation, in addition to the spacecraft hardware and launch vehicle selection and integration.
- The final design will be presented in the form of a written proposal document.

Learning Resources, Technology & Texts

Required Text:

Space Mission Engineering: The New SMAD (SME-SMAD) or just “SMAD”

Microcosm Press, Wertz/Everett/Puschell
978-1881883159, August 2011, 1033 pages

Available for $ 130 (USD) directly from the publisher:
https://astrobooks.com/spacemissionengineeringthenewsmadsme-smadwertzeverettpuschellavailablespring2011softcover.aspx

This is a standard reference, compiling much of the practical knowledge needed to design space systems, including key information on subsystems. This is an essential reference to own if you intend to work in the space field after graduation.

Additional Readings: There will be additional readings for some of the sub-sections. All will be distributed through links to the specific module on the Brightspace page when appropriate. Each module will include at least one reading from either a chapter in SMAD or a provided document.

Software/web resources: Some recommended software will be provided through Brightspace. For certain sub-system design problems, students are expected to determine and source the appropriate software tool, either provided by the university, available as open source, or developed on their own. *Selection of the appropriate tools (including software) is an important part of the design process.* Links to suggested tools will be provided on Brightspace. Students are expected to have access to and be familiar with the standard software used in the engineering industry, including word processors, spreadsheet editors, presentation designers (e.g. PowerPoint) and numeric computation tools (e.g., MATLAB). Students are free to choose the software and operating system(s) that work best for their project team. Teams should set up a common cloud storage area (e.g. OneDrive or Google Drive) to share work among members. **Students may not use any software that is subject to export control ("ITAR") regulations.**

Brightspace learning management system (LMS)
Access the course via Purdue's Brightspace learning management system. It is strongly suggested that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Learning Outcomes

On completing this course, students shall be able to:

1. Learn structured methods to design complex systems meeting customer requirements. Apply these methods to the design of a space system.
2. Understand basic models for typical spacecraft sub-systems (e.g. attitude control, communications, propulsion, etc.). Demonstrate the application of these models to assisting in preliminary design decisions and for sub-system sizing.
3. Develop a cost estimate, budget, schedule, and risk assessment for a space system design.
4. Give oral presentations and write technical reports and proposals.

Assignments

Design Project/Proposal:
The entire curriculum of AAE450 is based around a team project, intended to be conducted under the same conditions as an actual proposal development by professional space systems engineers. The final deliverables for the course are a

final presentation of up to one hour in length, and a comprehensive final report that complies with the format and selected requirements of an actual NASA mission proposal. The team's numerical score on the final presentation will comprise 15% of each team member's grade. The team's numerical score on the final proposal will comprise 35% of each team member's grade.

Final designs will be assessed based on:

- Requirements flow down
- Mission design
- Critical subsystems
- Cost and schedule estimation
- Risk analysis
- Presentation/ Proposal writing

Milestone Reviews

Throughout the semester, reviews will be held to monitor progress and offer feedback. During each review, each team will give an overview (up to one hour in length) of the state of their design. The initial review reflects the earliest phases of design, and for a real spacecraft design, it could be called the System Requirements Review (SRR) or the Conceptual Design Review (ConDR). The initial review is not graded and is intended primarily for teams to receive feedback and identify any problems with their design early on, allowing ample time to correct them.

Near the end of the semester, each team will perform a dress-rehearsal of the Final Review (possibly thought of as the "PDR" or Preliminary Design Review) to provide a final opportunity to discover and correct design flaws before their final presentation and report.

Note, attendance at all reviews is <u>mandatory</u> for all team members.

The instructors, and possibly from time to time, additional scientists and engineers invited as guests, will perform the review evaluations. The evaluators will review the final proposal (PDR) as if it were an actual submission seeking funding. The instructor will base the final numerical score for each design project on these reviews. A general rubric, derived from the stated proposal criteria, will be provided.

In addition to a request for detailed, specific feedback, the committee will be asked to answer the following question:

> *"If you were to receive this proposal in response to an actual announcement of opportunity, would you recommend it for selection and funding?"*

One criterion for a final score of "A" (90% or above) on the final project proposal is an affirmative response to this question by a majority of the review committee.

Weekly Lab Presentations and Module Questions

Each team will present every other week at their assigned lab presentation session. The purpose of these presentations is to update the instructor(s) on the progress of each team. The presentations should each be no more than 40 minutes in length and ideally should cover:

- A summary of the status of their design
- Results of analyses impacting the design
- Responses to weekly module questions
- Open action items requiring guidance from the instructor and the TAs.

The weekly module questions are a combination of readings and associated problems related to your team's design. Though ungraded, they are intended to provide a minimal set of action items to guide each team through the design process and ensure that focus is applied to the appropriate mission elements at the appropriate time. It is extremely important you address these questions in your bi-weekly update presentations, as it allows misunderstandings or mistakes to be caught early before they can propagate through the rest of your spacecraft design.

Because the lab presentations typically require only 1.5 – 2 hours of each lab period, the remaining hour of each lab session is available for groups to meet with each other and the instruction team.

Weekly Team Meetings

In addition to the weekly presentations, each team must have a weekly meeting to which all team members, the TA's, and the instructor are invited. This can be the group's regular meeting time, or a separate meeting. The instructor may not attend each of these meetings, but at least one of the TAs always will—in lieu of scheduled office hours. The purpose is to provide a regular work session with the TAs. If a team is struggling with understanding the lecture material, or has identified a difficult design problem, the TAs can suggest ways to solve it. Teams are encouraged to have further meetings each week focused on particular subsets of the spacecraft design. For example, an attitude control subsystem working group meeting with 2-4 members.

Teams will be formed partly by student preference stated during an early survey, and partly by the instructors. Teams will typically have 10-13 members.

Individual Attendance

Space missions have failed because critical personnel were absent from important decision meetings. *Attendance will therefore be taken at each lab presentation session in which your team presents.* Each student is allowed two (2) unexcused absences from these meetings, after which each further unexcused absence will result in a 5% reduction in course grade up to a maximum of 10%. *No unexcused absences are permitted for milestone reviews or the final PDR presentation*. Missing either without an excuse will result in a 5% deduction regardless of how many unexcused absences the student has already had. Students requesting an excused absence need to email the TA's and copy the instructor.

Individual Presentations

Each student must present for at least 10 minutes at one of their assigned lab presentation sessions during the semester. You should present on an analysis that you recently conducted to support the design. A percent grade will be awarded out of 5% maximum for this assignment.

Individual Project Contributions

Students will be scored on their personal contributions to the final spacecraft design and proposal. Contributions will be assessed by the instructor and the TAs. To support this assessment, the following documentation must be provided:

- Section authorship: Each section of the final report must have an author listed who was responsible for writing that section and obtaining the majority of its results. A single co-author may be optionally listed if another person contributed significantly.
- Weekly time sheets: Each team member must submit a weekly report of their hours (rounded to the nearest quarter hour) and which tasks they worked on.
- Team update sheets: Each team must fill out and submit a weekly update summarizing which meetings were conducted, who was present, and an account of all open/completed tasks. The update must conform to a template provided by the instructor and be signed off by the team's project manager and system engineer.

Individual Peer Review
An end of semester anonymous peer review will be provided. Students will assess the strengths and weakness of their group as well as the contributions of their peers.

Graded Assignments Summary
The grade breakdown is as follows:

Group Final Report	35%
Group Final Presentation	15%
Individual Attendance	10%
Individual Contributions	20%
Individual Peer Review	10%
Individual ABET Survey Completion	5%
Individual Presentation	5%

Grading Scale

Grading Philosophy
Final grades should measure the success of the team in applying material taught in the course, along with knowledge of the core AAE curriculum, to a practical design project. Each project will be evaluated on its own, not relative to the other teams. To repeat the statement given earlier, the essential question asked in the final review will be:

> *"If you were to receive this proposal in response to an actual announcement of opportunity, would you recommend it for selection and funding?"*

The main criterion for a final score of "A" (90% or above) is an affirmative response to this question by a majority of the review committee. The final numerical score for each student, computed from the combination of their team score along with their individual contributions described previous section, will be directly converted to a letter grade according to the intervals in the following list of final score percentages:

A: ≥ 90
B: $[80, 90)$
C: $[70, 80)$
D: $[60, 70)$
F: < 60